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Scitech Institutions in India

The story of developing India's research infrastructure

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n his book *Battles of the Honorable East India Company*, Wing Commander (retd.) Dr. M.S.

Naravane identifies a few major factors that contributed to the British triumphing in India in the first place. These factors include a major gap in education, technological advancement, and industrialisation. The officers of the Company were well educated, but also studied the local history and culture, and took pains to become fluent in local languages. It was only a few of the local leaders on the other side who were literate. In terms of technologies available at the grassroots level, the first printing press came to India almost a century after books became widespread in the West, and similarly, mechanical clocks in India were novelty items centuries after they became common in Europe.

The industrial revolution that spread like wildfire in the west was slow to arrive in India. While Hyder Ali and his son Tipu Sultan did manage to surprise the British (and their horses) with their ironclad war rockets, the artillery of the British was considered far superior, at least during the early phases of the conflict.



A rocket being lit by a Mysorean soldier

The local leaders started establishing their own gun factories, starting with Benoît de Boigne opening up a canon foundry in Agra in the 1780s. The Europeans had been careful about selling inferior goods to the local Indians, but these local foundries produced weapons as good as any, but by then it was too late.

In the 19th Century, under the Raj, there were a number of local educational institutions set up in the country, so that the "natives" could learn the subjects of agriculture, engineering and medicine, all of which were considered important for the advancement of British objectives in India. In 1857, there were universities established in Calcutta, Bombay and Madras, but these were all aimed for

training and recruiting locals into the civil service, and offered few scientific or technical courses. Around this time, there were also societies formed for the specific study of geology, geography, botany, zoology and other specialised branches of science. Students had access to scientific collections in museums, botanical gardens and libraries. The aristocracy of India contributed to the setting up of institutions for technical education, both traditional and modern.

Rajiva Bhatnagar, an ex BARC scientist tells the story of how science came to India with colonial powers in the book *Whoduunit - Pathology of Modern Indian Science*. The doctors from the medical institutions became amateur botanists and zoologists